

In randomly generating 10k clusters, sizes 50-150 atoms, of the following elements: Al, Co, Mn, Ni, W of randomly chosen compositions, relaxed fully to 0.05 eV/Ang with both the MACE foundational r2SCAN model as well as that same model fine-tuned on DFT calculations, the following trends were found.

Formation energies were calculated with both models using the following formula:

$$E_{form} = \frac{1}{N}(E_{cluster} - E_{mixed\ bulk}) = EPA - \sum \mu_i Z_i$$

Where EPA is the energy per atom, where N is the atoms per cluster, and the sum adds up the bulk reference energy per atom μ , scaled by its compositional fraction per cluster z .

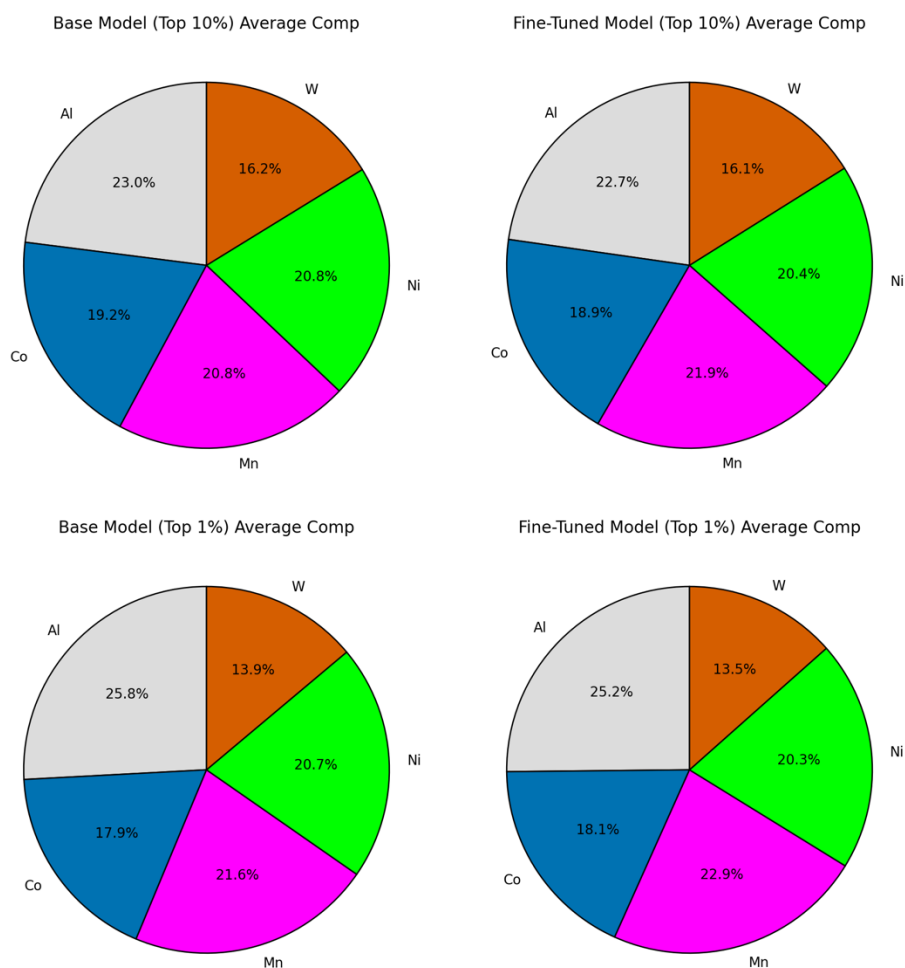


Figure 1: Elemental ratios of the top (most stable) 10% above and top 1% below, representing 1000 and 100 clusters respectively. We note that the trend is Al>Mn>Ni>Co>W.

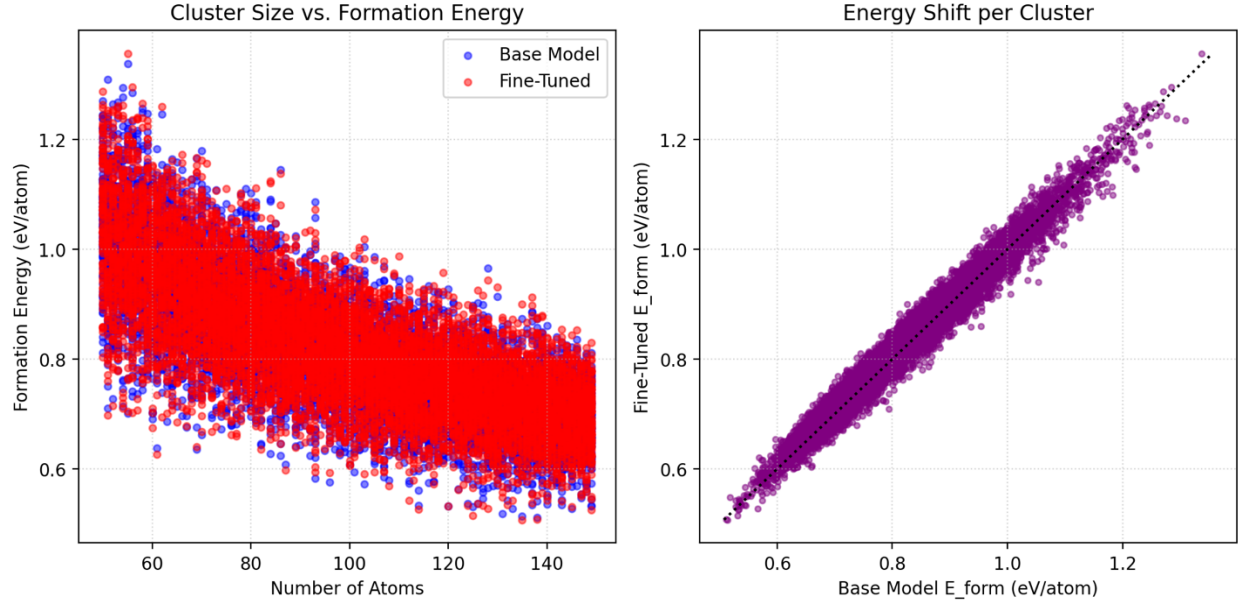


Figure 2. E_{form} as a function of cluster size (left), and E_{form} predictions compared between base model and fine-tuned model (right). Size appears to play a negative power-law contribution to E_{form} , which should indeed tend to the zero point ($E_{mixed\ bulk}$). Fitting a power law equation, we find $N^{-0.457}$ for the base model and $N^{-0.356}$ for the fine-tuned model (similar to the surface area per atom $\propto N^{-0.333}$)

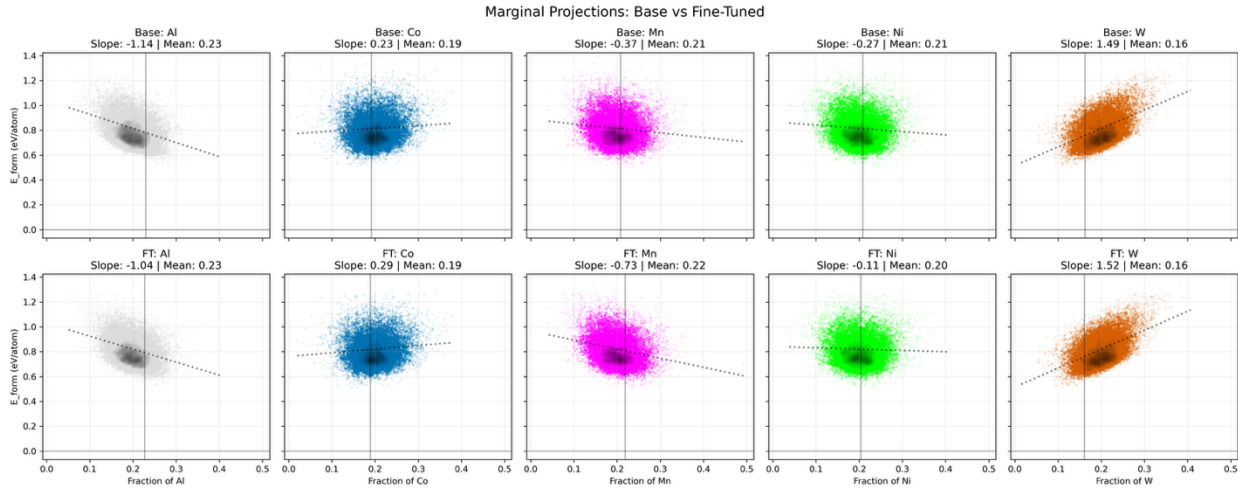


Figure 3. E_{form} statistics (Gaussian KDEs) for the whole 10k clusters, as a function of their elemental fractions, for both the base model (top) and the fine-tuned model (bottom). Means and slopes are reported. Note that only Al and Mn have negative slopes, which indicates loosely, that more is better.

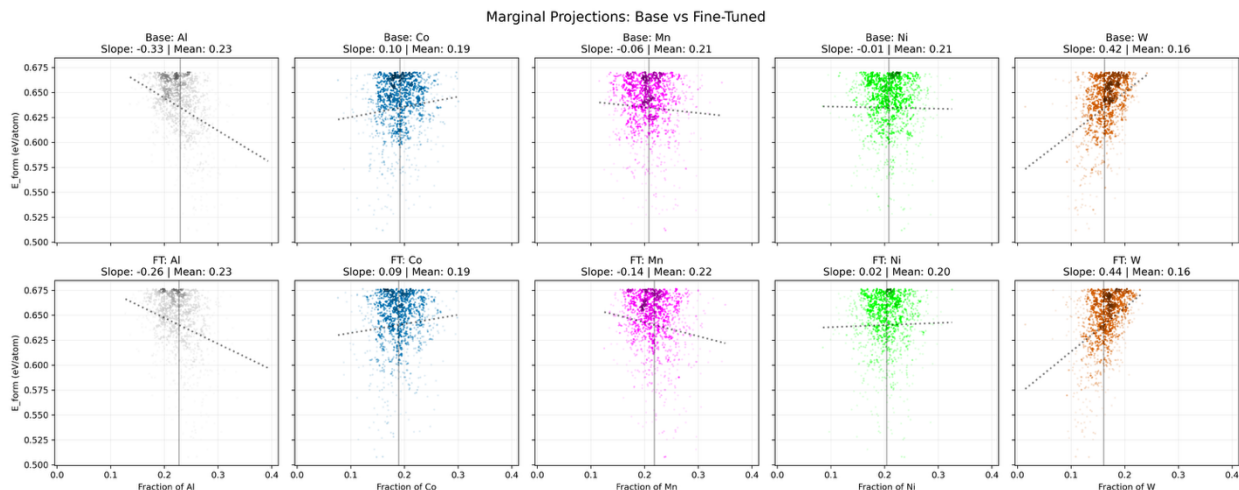


Figure 4. Same plot as above, this time exclusively for the top 10%. Note that here, the y axis starts at 0.5 rather than at 0.0, and only goes up to 0.7 rather than 1.4. This graph emphasizes the previous trends, namely that Al is highly favored, and W is highly disfavored, while Mn is somewhat favored, and Co somewhat disfavored – Ni is relatively ambivalent.

A GA over the set of possible compositions was run, where clusters had a fixed size of either 55 or 147 atoms. Results quickly saturated after 5 generations, some are shown below, where the elemental hierarchy as a function of E_{form} now becomes Al>Ni>Mn>Co>W.

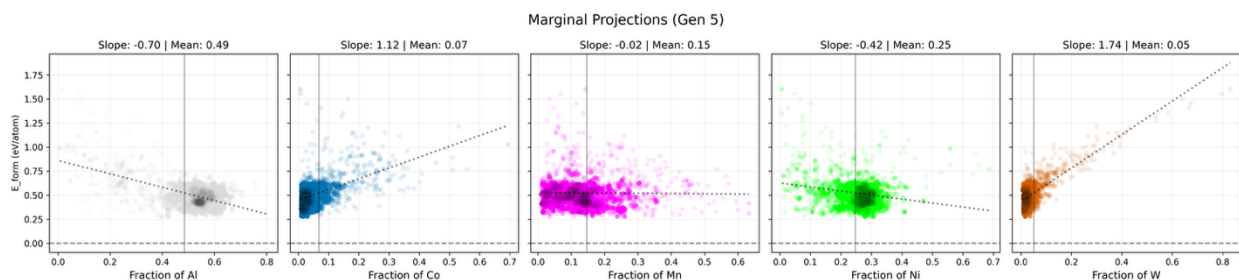
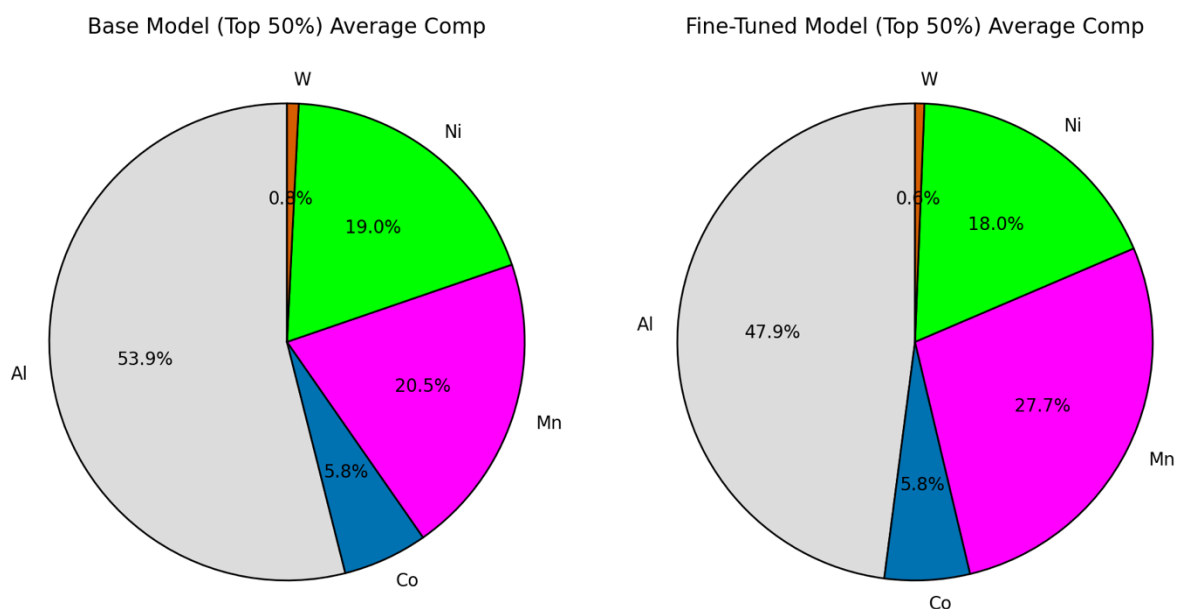


Figure 5. Notice Mn and Ni have switched places in the E_{form} percentage hierarchy, relative to the random structures, yet simultaneously, we notice Ni does not like to extend beyond 50%.

These differences are likely because now we are looking at top performing composition ratios individually, rather than looking at their statistics. Note as well, we've removed all data points wherein a given elemental fraction is reduced to zero, leaving around 3k compositions. However, these GA compositions are simulated with clusters sliced neatly out of bulks, rather than random or amorphous in geometry. Hence, some of the top performing compositions have been extracted and simulated similarly to the random ones, with random sizes again between 50 and 150 atoms.

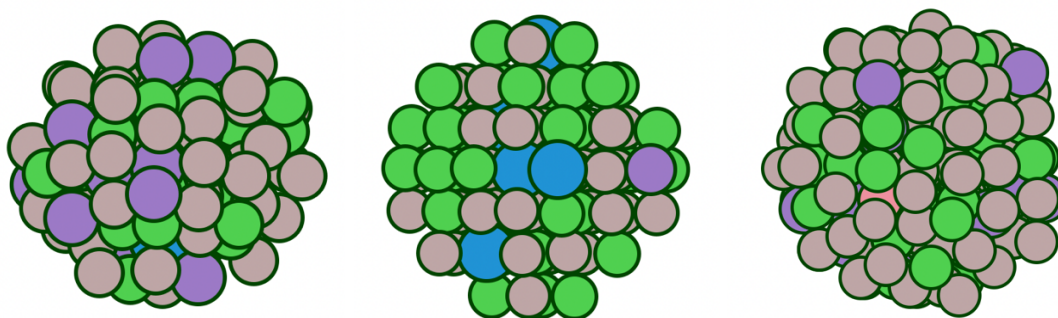
Results are shown below.

Results are still coming in, but from the top compositions suggested by the GA, we see the hierarchy remains as it was for the random structures, thereby indicating that the GA's geometries as cut from bulk structures skewed the rankings of E_{form} . We also notice the base model and the fine-tuned model disagree as they did in Figure 1, yet their disagreement here has been emphasized – the base model prefers more Al, whereas the fine-tuned model prefers more Mn, with the other elemental ratios remaining in very reasonable agreement. Note as well that we keep the top 50%, around 500 clusters, because they already represent a specific subregion of the possible composition space, as deemed stable by the GA.



More DFT results are coming in and continue to improve the fine-tuned model's predictive accuracy on new DFT clusters.

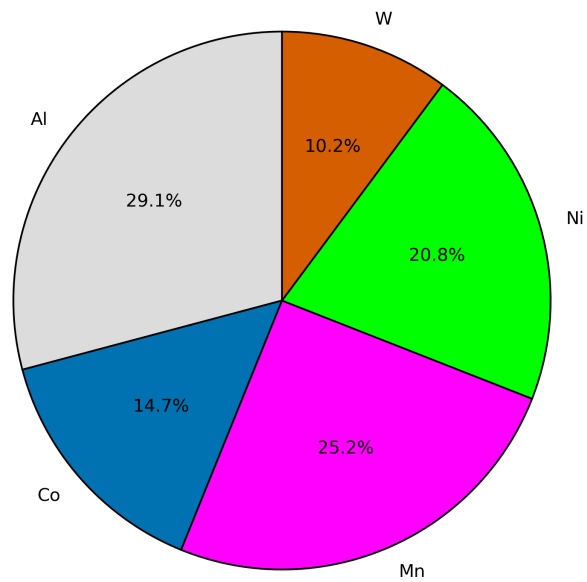
Reference images: (left to right) Randomized relaxed, GA cut from bulk, GA relaxed.



New DFT results.

Randomized 55 atom clusters:

Top 10% Most Stable Clusters
Average Composition



Marginal Projections (DFT-values)

